

YAMPA VALLEY, CO, USA

WMO: 725710

Lat: 40.481N

Lon: 107.218W

Elev: 6600

StdP: 11.51

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94025

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	0.3	-15.0	3.1	-4.0	-7.5	4.7	0.9	24.8	31.1	20.9	27.1	5.7	110	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.5	88.1	56.3	84.7	55.5	82.3	55.0	59.6	78.1	58.5	77.1	57.2	76.0	10.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.3	80.4	63.2	52.3	74.7	63.1	50.0	68.4	63.8	29.3	78.3	28.3	77.2	27.5	76.1	66.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.9	19.8	17.2		DB	-14.7	91.6	5.6	2.4	-18.7	93.4	-21.9	94.8	-25.1	96.2	-29.1	97.9
				WB	-14.8	61.5	5.3	1.8	-18.6	62.8	-21.7	63.8	-24.7	64.8	-28.5	66.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	19.6	23.6	33.1	41.9	50.9	60.9	68.4	66.1	58.0	45.0	33.0	20.8	
	DBStd	18.61	9.44	8.97	9.02	7.86	7.18	6.53	4.17	3.94	6.77	8.63	9.62	9.69	
	HDD50	4180	944	740	527	256	78	6	0	0	17	196	511	905	
	HDD65	8064	1409	1160	990	692	438	151	17	34	221	621	961	1370	
	CDD50	1820	0	0	2	14	106	333	570	498	256	40	1	0	
	CDD65	227	0	0	0	0	1	27	122	67	10	0	0	0	
	CDH74	4135	0	0	0	1	55	782	1817	1143	332	5	0	0	
	CDH80	1210	0	0	0	0	5	210	644	299	52	0	0	0	
Wind	WSAvg	7.9	6.9	7.0	7.7	8.7	8.4	8.6	8.2	8.3	8.3	7.8	7.3	6.9	
Precipitation	PrecAvg	20.4	1.9	1.6	1.5	2.1	1.7	1.3	1.4	1.6	2.2	1.8	1.8	1.5	
	PrecMax	28.5	4.2	3.0	2.5	3.7	4.5	3.8	3.0	3.3	5.9	3.5	3.0	2.6	
	PrecMin	14.0	0.7	0.9	0.6	1.3	0.5	0.3	0.4	0.4	1.0	0.4	0.7	0.7	
	PrecStd	3.3	0.9	0.6	0.6	0.7	0.9	0.8	0.6	0.6	1.1	0.8	0.6	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	46.1	64.4	71.4	79.6	89.8	91.1	88.5	84.5	73.4	63.5	45.9	
		MCWB	34.0	35.9	43.3	47.3	52.3	56.1	57.3	56.3	54.8	48.8	43.3	35.8	
	2%	DB	37.3	42.6	57.2	66.0	74.8	84.7	89.5	85.9	81.3	70.2	57.2	39.4	
		MCWB	31.6	34.0	40.4	44.7	50.5	54.4	57.1	55.5	53.0	47.8	40.5	32.7	
	5%	DB	35.9	38.7	52.3	63.0	72.0	81.9	86.4	83.4	78.7	65.7	53.5	36.6	
		MCWB	31.4	32.3	38.6	43.1	49.5	53.8	56.3	55.1	51.9	46.2	39.4	31.1	
	10%	DB	33.5	36.2	46.3	57.5	67.7	79.1	83.7	81.1	74.5	61.3	47.6	34.1	
		MCWB	30.1	31.1	36.2	41.2	48.4	53.1	56.1	54.7	51.0	43.9	37.1	29.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.3	37.0	44.2	48.1	54.6	58.5	61.5	60.5	58.3	51.5	44.5	36.6
			MCDB	39.9	45.1	62.5	67.6	75.1	80.5	79.3	76.5	76.5	68.0	60.0	44.9
2%		WB	33.0	34.7	41.5	45.6	52.2	56.8	60.1	59.1	55.6	49.0	41.7	33.7	
		MCDB	36.7	40.8	56.0	63.1	70.9	79.0	78.9	76.1	73.1	65.9	55.6	38.9	
5%		WB	31.5	33.0	39.1	43.9	50.5	55.3	59.1	57.7	54.0	47.0	39.7	31.9	
		MCDB	35.0	38.0	50.8	60.4	68.1	77.8	78.3	75.1	70.7	63.1	51.8	36.0	
10%		WB	29.4	31.5	36.7	42.1	48.8	54.0	58.0	56.6	52.6	45.1	37.3	29.9	
		MCDB	32.5	35.6	46.2	56.7	65.7	75.3	77.0	74.4	69.6	59.8	46.3	33.5	
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.6	19.7	22.5	24.9	30.0	29.5	28.4	27.8	24.4	20.4	16.2	
		MCDBR	16.6	18.4	27.2	30.3	31.9	34.5	33.1	32.6	33.0	32.1	28.4	17.7	
		MCWBR	12.6	12.6	14.2	14.2	14.0	13.8	11.8	11.9	13.5	15.6	15.5	12.5	
	5% WB	MCDBR	15.4	16.4	25.1	27.9	28.8	31.2	27.7	26.2	26.8	28.2	26.2	16.3	
		MCWBR	12.3	11.6	13.6	13.5	12.9	13.0	10.3	10.1	11.4	13.9	14.7	11.7	
Clear-Sky Solar Irradiance	taub	0.230	0.246	0.268	0.311	0.315	0.320	0.347	0.343	0.295	0.266	0.248	0.239		
	taud	2.454	2.419	2.449	2.404	2.478	2.461	2.433	2.440	2.582	2.638	2.570	2.455		
	Ebn at Noon	305	313	316	305	305	302	292	292	302	304	297	292		
	Edh at Noon	24	29	32	36	34	35	36	34	28	23	21	22		
All-Sky Solar Radiation	RadAvg	705	967	1406	1733	2001	2409	2242	1934	1643	1151	770	584		
	RadStd	44	71	128	123	148	144	96	87	105	97	85	51		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page